

LAPORAN PRAKTIKUM BASIS DATA II
PERTEMUAN 13: STORED PROCEDURE, TRIGGER DAN VIEW



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PROGRAM STUDI TEKNIK INFORMATIKA

POLITEKNIK PURBAYA

TEGAL

2026

I. Tujuan Praktikum

1. Memahami Stored Procedure.
2. Memahami Trigger.
3. Memahami View.
4. Mengimplementasikan ketiganya pada database sidang.

II. Dasar Teori

Stored Procedure adalah kumpulan perintah SQL yang disimpan di server database dan dapat dipanggil kembali. Trigger adalah program yang berjalan otomatis ketika terjadi INSERT, UPDATE, atau DELETE. View adalah tabel virtual yang berasal dari hasil query satu atau beberapa tabel.

III. Langkah Praktikum

1. Masuk ke MariaDB melalui CMD.

```
Command Prompt - mysql -u x + v
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ADVAN>cd C:\xampp1\mysql\bin

C:\xampp1\mysql\bin>mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 24
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

2. Membuat database db_sidang.

```
MariaDB [(none)]> CREATE DATABASE db_sidang;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> USE db_sidang;
Database changed
MariaDB [db_sidang]> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| db_sidang  |
+-----+
1 row in set (0.000 sec)

MariaDB [db_sidang]>
```

3. Membuat tabel

Membuat tabel mahasiswa, dosen, ruangan, sidang, penguji, dan log_activity.

```
MariaDB [db_sidang]> CREATE TABLE mahasiswa (  
-> id_mahasiswa INT AUTO_INCREMENT PRIMARY KEY,  
-> nim VARCHAR(20) NOT NULL,  
-> nama VARCHAR(100) NOT NULL,  
-> prodi VARCHAR(100) NOT NULL  
-> );  
Query OK, 0 rows affected (0.014 sec)  
  
MariaDB [db_sidang]>  
MariaDB [db_sidang]> CREATE TABLE dosen (  
-> id_dosen INT AUTO_INCREMENT PRIMARY KEY,  
-> nama_dosen VARCHAR(100) NOT NULL,  
-> jabatan VARCHAR(100)  
-> );  
Query OK, 0 rows affected (0.010 sec)  
  
MariaDB [db_sidang]>  
MariaDB [db_sidang]> CREATE TABLE ruangan (  
-> id_ruangan INT AUTO_INCREMENT PRIMARY KEY,  
-> nama_ruangan VARCHAR(50) NOT NULL  
-> );  
Query OK, 0 rows affected (0.010 sec)  
  
MariaDB [db_sidang]>  
MariaDB [db_sidang]> CREATE TABLE sidang (  
-> id_sidang INT AUTO_INCREMENT PRIMARY KEY,  
-> id_mahasiswa INT,  
-> tanggal_sidang DATE,  
-> jam_mulai TIME,  
-> jam_selesai TIME,  
-> id_ruangan INT,  
-> FOREIGN KEY (id_mahasiswa) REFERENCES mahasiswa(id_mahasiswa),  
-> FOREIGN KEY (id_ruangan) REFERENCES ruangan(id_ruangan)  
-> );  
Query OK, 0 rows affected (0.029 sec)  
  
MariaDB [db_sidang]>  
MariaDB [db_sidang]> CREATE TABLE penguji (  
-> id_penguji INT AUTO_INCREMENT PRIMARY KEY,  
-> id_sidang INT,  
-> id_dosen INT,  
-> peran VARCHAR(50),  
-> FOREIGN KEY (id_sidang) REFERENCES sidang(id_sidang),  
-> FOREIGN KEY (id_dosen) REFERENCES dosen(id_dosen)  
-> );  
Query OK, 0 rows affected (0.028 sec)  
  
MariaDB [db_sidang]>  
MariaDB [db_sidang]> CREATE TABLE log_activity (  
-> id_log INT AUTO_INCREMENT PRIMARY KEY,  
-> aktivitas VARCHAR(255),  
-> waktu DATETIME  
-> );  
Query OK, 0 rows affected (0.036 sec)
```

```
MariaDB [db_sidang]> SHOW TABLES;
```

```
+-----+  
| Tables_in_db_sidang |  
+-----+  
| dosen                |  
| log_activity         |  
| mahasiswa            |  
| penguji              |  
| ruangan              |  
| sidang               |  
+-----+
```

```
6 rows in set (0.001 sec)
```

```
MariaDB [db_sidang]>
```

4. Melakukan insert data.

```
MariaDB [db_sidang]> INSERT INTO mahasiswa (nim, nama, prodi)
-> VALUES
-> ('2302001','Dwi Sekar Arum','Teknik Informatika'),
-> ('2302002','Devi Kurnia','Teknik Informatika'),
-> ('2302003','Mega Ario','Teknik Informatika'),
-> ('2302004','Sri Zakiya','Teknik Informatika'),
-> ('2302005','Zahwa Amir','Teknik Informatika');
Query OK, 5 rows affected (0.027 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [db_sidang]> INSERT INTO dosen (nama_dosen, jabatan)
-> VALUES
-> ('Muhamad Ainurohman, S.Kom','Ketua Penguji'),
-> ('Denny Vasanando Sabanise, S.Kom., M.Kom','Penguji 1'),
-> ('Fitriasih, S.Kom., M.Kom','Penguji 2');
Query OK, 3 rows affected (0.014 sec)
Records: 3 Duplicates: 0 Warnings: 0

MariaDB [db_sidang]> INSERT INTO ruangan (nama_ruangan)
-> VALUES
-> ('Ruang A'),
-> ('Ruang B'),
-> ('Ruang C');
Query OK, 3 rows affected (0.048 sec)
Records: 3 Duplicates: 0 Warnings: 0

MariaDB [db_sidang]>
```

```
MariaDB [db_sidang]> SELECT * FROM mahasiswa;
+-----+-----+-----+-----+
| id_mahasiswa | nim      | nama           | prodi           |
+-----+-----+-----+-----+
| 1             | 2302001 | Dwi Sekar Arum | Teknik Informatika |
| 2             | 2302002 | Devi Kurnia    | Teknik Informatika |
| 3             | 2302003 | Mega Ario      | Teknik Informatika |
| 4             | 2302004 | Sri Zakiya     | Teknik Informatika |
| 5             | 2302005 | Zahwa Amir    | Teknik Informatika |
+-----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [db_sidang]> SELECT * FROM dosen;
+-----+-----+-----+
| id_dosen | nama_dosen                                | jabatan |
+-----+-----+-----+
| 1        | Muhamad Ainurohman, S.Kom                | Ketua Penguji |
| 2        | Denny Vasanando Sabanise, S.Kom., M.Kom | Penguji 1 |
| 3        | Fitriasih, S.Kom., M.Kom                | Penguji 2 |
+-----+-----+-----+
3 rows in set (0.001 sec)

MariaDB [db_sidang]> SELECT * FROM ruangan;
+-----+-----+
| id_ruangan | nama_ruangan |
+-----+-----+
| 1          | Ruang A      |
| 2          | Ruang B      |
| 3          | Ruang C      |
+-----+-----+
3 rows in set (0.001 sec)

MariaDB [db_sidang]>
```

5. Membuat Stored Procedure.

Penjelasan: Membuat Stored Procedure.

```
MariaDB [db_sidang]> DELIMITER $$
MariaDB [db_sidang]>
MariaDB [db_sidang]> CREATE PROCEDURE sp_tambah_sidang(
  -> IN p_id_mahasiswa INT,
  -> IN p_tanggal DATE,
  -> IN p_jam_mulai TIME,
  -> IN p_jam_selesai TIME,
  -> IN p_id_ruangan INT
  -> )
  ->
  -> BEGIN
  ->
  -> INSERT INTO sidang(
  -> id_mahasiswa,
  -> tanggal_sidang,
  -> jam_mulai,
  -> jam_selesai,
  -> id_ruangan
  -> )
  ->
  -> VALUES(
  -> p_id_mahasiswa,
  -> p_tanggal,
  -> p_jam_mulai,
  -> p_jam_selesai,
  -> p_id_ruangan
  -> );
  ->
  -> END $$
Query OK, 0 rows affected (0.047 sec)

MariaDB [db_sidang]>
MariaDB [db_sidang]> DELIMITER ;
MariaDB [db_sidang]>
```

6. Menguji Stored Procedure.

```
MariaDB [db_sidang]>
MariaDB [db_sidang]> DELIMITER ;
MariaDB [db_sidang]> CALL sp_tambah_sidang(
  -> 1,
  -> '2026-07-10',
  -> '08:00:00',
  -> '09:30:00',
  -> 1
  -> );
Query OK, 1 row affected (0.023 sec)

MariaDB [db_sidang]>
MariaDB [db_sidang]> CALL sp_tambah_sidang(
  -> 2,
  -> '2026-07-10',
  -> '10:00:00',
  -> '11:30:00',
  -> 2
  -> );
Query OK, 1 row affected (0.015 sec)

MariaDB [db_sidang]>
MariaDB [db_sidang]> CALL sp_tambah_sidang(
  -> 3,
  -> '2026-07-11',
  -> '08:00:00',
  -> '09:30:00',
  -> 3
  -> );
Query OK, 1 row affected (0.011 sec)

MariaDB [db_sidang]> |
```

7. Hasil Procedure

```
MariaDB [db_sidang]> SELECT * FROM sidang;
```

id_sidang	id_mahasiswa	tanggal_sidang	jam_mulai	jam_selesai	id_ruangan
1	1	2026-07-10	08:00:00	09:30:00	1
2	2	2026-07-10	10:00:00	11:30:00	2
3	3	2026-07-11	08:00:00	09:30:00	3

```
3 rows in set (0.001 sec)

MariaDB [db_sidang]>
```

8. Membuat View dan menguji hasilnya.

```
MariaDB [db_sidang]> CREATE VIEW v_jadwal_sidang AS
->
-> SELECT
-> m.nim,
-> m.nama,
-> m.prodi,
-> s.tanggal_sidang,
-> CONCAT(s.jam_mulai, ' - ', s.jam_selesai) AS jam_sidang,
-> r.nama_ruangan
-> FROM sidang s
->
-> JOIN mahasiswa m
-> ON s.id_mahasiswa=m.id_mahasiswa
->
-> JOIN ruangan r
-> ON s.id_ruangan=r.id_ruangan;
Query OK, 0 rows affected (0.057 sec)

MariaDB [db_sidang]>
```

```
MariaDB [db_sidang]> SELECT * FROM v_jadwal_sidang;
```

nim	nama	prodi	tanggal_sidang	jam_sidang	nama_ruangan
2302001	Dwi Sekar Arum	Teknik Informatika	2026-07-10	08:00:00 - 09:30:00	Ruang A
2302002	Devi Kurnia	Teknik Informatika	2026-07-10	10:00:00 - 11:30:00	Ruang B
2302003	Mega Ario	Teknik Informatika	2026-07-11	08:00:00 - 09:30:00	Ruang C

```
3 rows in set (0.002 sec)

MariaDB [db_sidang]>
```

9. Membuat Trigger insert

```
MariaDB [db_sidang]> DELIMITER $$
MariaDB [db_sidang]>
MariaDB [db_sidang]> CREATE TRIGGER trg_log_sidang
->
-> AFTER INSERT
-> ON sidang
->
-> FOR EACH ROW
->
-> BEGIN
->
-> INSERT INTO log_activity(
-> aktivitas,
-> waktu
-> )
->
-> VALUES(
-> CONCAT('Tambah sidang mahasiswa ID ', NEW.id_mahasiswa),
-> NOW()
-> );
->
-> END $$
Query OK, 0 rows affected (0.030 sec)

MariaDB [db_sidang]>
MariaDB [db_sidang]> DELIMITER ;
MariaDB [db_sidang]>
```

```

MariaDB [db_sidang]>
MariaDB [db_sidang]> DELIMITER ;
MariaDB [db_sidang]> CALL sp_tambah_sidang(
-> 4,
-> '2026-07-12',
-> '13:00:00',
-> '14:30:00',
-> 1
-> );
Query OK, 2 rows affected (0.016 sec)

MariaDB [db_sidang]> SELECT * FROM log_activity;
+-----+-----+-----+
| id_log | aktivitas | waktu |
+-----+-----+-----+
| 1 | Tambah sidang mahasiswa ID 4 | 2026-07-02 14:50:34 |
+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [db_sidang]>

```

10. Membuat Trigger delete.

```

MariaDB [db_sidang]> DELIMITER $$
MariaDB [db_sidang]>
MariaDB [db_sidang]> CREATE TRIGGER trg_hapus_sidang
->
-> AFTER DELETE
-> ON sidang
->
-> FOR EACH ROW
->
-> BEGIN
->
-> INSERT INTO log_activity(
-> aktivitas,
-> waktu
-> )
->
-> VALUES(
-> CONCAT('Hapus sidang mahasiswa ID ', OLD.id_mahasiswa),
-> NOW()
-> );
->
-> END $$
Query OK, 0 rows affected (0.021 sec)

MariaDB [db_sidang]>
MariaDB [db_sidang]> DELIMITER ;
MariaDB [db_sidang]>

```

```

-> END $$
Query OK, 0 rows affected (0.021 sec)

MariaDB [db_sidang]>
MariaDB [db_sidang]> DELIMITER ;
MariaDB [db_sidang]> DELETE FROM sidang
-> WHERE id_sidang=4;
Query OK, 1 row affected (0.013 sec)

MariaDB [db_sidang]> SELECT * FROM log_activity;
+-----+-----+-----+
| id_log | aktivitas | waktu |
+-----+-----+-----+
| 1 | Tambah sidang mahasiswa ID 4 | 2026-07-02 14:50:34 |
| 2 | Hapus sidang mahasiswa ID 4 | 2026-07-02 14:58:34 |
+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [db_sidang]>

```

11. Membuat Query laporan.

```

MariaDB [db_sidang]> SELECT *
-> FROM mahasiswa
-> WHERE id_mahasiswa NOT IN
-> (
-> SELECT id_mahasiswa
-> FROM sidang
-> );
+-----+-----+-----+
| id_mahasiswa | nim | nama | prodi |
+-----+-----+-----+
| 5 | 2302005 | Zahwa Amir | Teknik Informatika |
+-----+-----+-----+
1 row in set (0.002 sec)

MariaDB [db_sidang]>

```

12. Membuat Procedure cek bentrok.

```
MariaDB [db_sidang]>
MariaDB [db_sidang]> CREATE PROCEDURE sp_tambah_sidang_cek(
->
-> IN p_id_mahasiswa INT,
-> IN p_tanggal DATE,
-> IN p_jam_mulai TIME,
-> IN p_jam_selesai TIME,
-> IN p_id_ruangan INT
->
-> )
->
-> BEGIN
->
-> IF EXISTS(
->
-> SELECT *
->
-> FROM sidang
->
-> WHERE tanggal_sidang=p_tanggal
->
-> AND id_ruangan=p_id_ruangan
->
-> AND(
->
-> (p_jam_mulai BETWEEN jam_mulai AND jam_selesai)
->
-> OR
->
-> (p_jam_selesai BETWEEN jam_mulai AND jam_selesai)
->
-> )
->
-> )
->
-> THEN
->
-> SELECT 'Jadwal Ruangan Bentrok' AS Pesan;
->
-> ELSE
->
-> INSERT INTO sidang(
->
-> id_mahasiswa,
-> tanggal_sidang,
-> jam_mulai,
-> jam_selesai,
-> id_ruangan
->
-> )
->
-> VALUES(
->
-> p_id_mahasiswa,
-> p_tanggal,
-> p_jam_mulai,
-> p_jam_selesai,
-> p_id_ruangan
->
-> );
->
-> END IF;
->
-> END $$
Query OK, 0 rows affected (0.026 sec)

MariaDB [db_sidang]>
MariaDB [db_sidang]> DELIMITER ;
MariaDB [db_sidang]>
```

```
MariaDB [db_sidang]> DELIMITER ;
MariaDB [db_sidang]> CALL sp_tambah_sidang_cek(
-> 5,
-> '2026-07-10',
-> '08:30:00',
-> '09:00:00',
-> 1
-> );
+-----+
| Pesan |
+-----+
| Jadwal Ruangan Bentrok |
+-----+
1 row in set (0.001 sec)

Query OK, 0 rows affected (0.003 sec)

MariaDB [db_sidang]>
```


13. Membuat View penguji.

```
MariaDB [db_sidang]> CREATE VIEW v_dosen_penguji AS
-> SELECT
-> s.id_sidang,
-> m.nama,
-> d.nama_dosen,
-> p.peran
-> FROM penguji p
-> JOIN dosen d
-> ON p.id_dosen=d.id_dosen
-> JOIN sidang s
-> ON p.id_sidang=s.id_sidang
-> JOIN mahasiswa m
-> ON s.id_mahasiswa=m.id_mahasiswa;
Query OK, 0 rows affected (0.067 sec)

MariaDB [db_sidang]>
```

```
MariaDB [db_sidang]> INSERT INTO penguji(id_sidang,id_dosen,peran)
-> VALUES
-> (1,1,'Ketua'),
-> (1,2,'Penguji 1'),
-> (1,3,'Penguji 2');
Query OK, 3 rows affected (0.010 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_sidang]> SELECT * FROM v_dosen_penguji;
```

id_sidang	nama	nama_dosen	peran
1	Dwi Sekar Arum	Muhamad Ainurohman, S.Kom	Ketua
1	Dwi Sekar Arum	Denny Vasanando Sabanise, S.Kom., M.Kom	Penguji 1
1	Dwi Sekar Arum	Fitriasih, S.Kom., M.Kom	Penguji 2

3 rows in set (0.002 sec)

```
MariaDB [db_sidang]>
```

14. Laporan jumlah sidang

```

MariaDB [db_sidang]>
MariaDB [db_sidang]> SELECT
->
-> tanggal_sidang,
->
-> COUNT(*) AS jumlah_sidang
->
-> FROM sidang
->
-> GROUP BY tanggal_sidang
->
-> ORDER BY tanggal_sidang;
+-----+-----+
| tanggal_sidang | jumlah_sidang |
+-----+-----+
| 2026-07-10     | 2             |
| 2026-07-11     | 1             |
+-----+-----+
2 rows in set (0.001 sec)

MariaDB [db_sidang]>

```

12. Mengekspor database menjadi db_sidang.sql.

```

C:\xampp1\mysql\bin>mysqldump -u root -p db_sidang > db_sidang.sql
Enter password:

C:\xampp1\mysql\bin>dir db_sidang.sql
Volume in drive C has no label.
Volume Serial Number is 222F-9701

Directory of C:\xampp1\mysql\bin

02/07/2026  21:54                12.502 db_sidang.sql
               1 File(s)              12.502 bytes
               0 Dir(s)  167.535.259.648 bytes free

C:\xampp1\mysql\bin>

```

IV. Analisis

Stored Procedure mempermudah otomatisasi proses penambahan jadwal sidang. View menyederhanakan penyajian data dari beberapa tabel sehingga lebih mudah dibaca. Trigger mencatat aktivitas database secara otomatis sehingga membantu proses audit data.

V. Kesimpulan

1. Stored Procedure mempermudah pengolahan data.
2. View memudahkan penyajian informasi.
3. Trigger membantu pencatatan aktivitas database secara otomatis.
4. Praktikum berhasil dilaksanakan dan database berhasil diekspor menjadi db_sidang.sql.